

BIGFORK, MN, USA

WMO: 727468

Lat: 47.783N

Lon: 93.650W

Elev: 411

StdP: 96.48

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-27.8	-35.9	0.1	-32.4	-32.2	0.2	-27.9	8.5	-6.1	7.6	-6.1	0.3	290	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 29.1	(d) 21.4	(e) 27.8	(f) 20.3	(g) 27.1	(h) 19.6	(i) 23.4	(j) 27.5	(k) 22.1	(l) 26.3	(m) 20.8	(n) 25.1	(o) 4.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.3	17.8	26.2	21.0	16.5	25.2	19.0	14.5	23.3	71.7	28.0	66.9	26.2	62.2	25.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.8	(b) 6.9	(c) 5.7	(e) -35.8	(f) 32.3	(g) 4.1	(h) 1.7	(i) -38.8	(j) 33.5	(k) -41.2	(l) 34.4	(m) -43.5	(n) 35.4	(o) -46.5	(p) 36.6		
(a) 7.8	(b) 6.9	(c) 5.7	(e) -35.7	(f) 24.7	(g) 3.7	(h) 1.4	(i) -38.4	(j) 25.7	(k) -40.6	(l) 26.6	(m) -42.7	(n) 27.3	(o) -45.5	(p) 28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-13.1	-11.6	-3.4	4.6	11.2	16.2	19.0	17.4	13.3	6.3	-1.6	-10.0		
	DBStd	12.66	7.96	7.49	7.74	5.41	4.70	3.68	3.26	3.48	4.69	4.98	6.31	7.53		
	HDD10.0	3087	717	604	417	176	42	2	0	1	19	139	351	622		
	HDD18.3	5316	975	837	673	413	228	83	31	59	162	375	599	880		
	CDD10.0	934	0	0	2	13	77	188	280	231	118	23	2	0		
	CDD18.3	122	0	0	0	1	6	20	53	31	11	1	0	0		
	CDH23.3	1435	0	0	1	7	91	241	614	362	109	10	0	0		
	CDH26.7	319	0	0	0	1	17	46	149	85	19	1	0	0		
(14) Wind	WSAvg	2.3	2.3	2.2	2.5	2.7	2.7	2.2	1.9	1.8	2.2	2.4	2.4	2.2		
(15) Precipitation	PrecAvg	703	20	16	27	50	77	103	110	83	84	70	37	26		
	PrecMax	903	51	36	83	125	161	204	254	142	196	139	84	41		
	PrecMin	541	4	1	6	23	27	29	50	25	22	5	1	6		
	PrecStd	86	13	9	17	26	32	39	50	36	36	32	24	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.7	18.0	23.0	28.8	30.5	32.0	32.0	28.9	24.2	17.0	6.0		
		MCWB	2.7	4.3	11.7	13.9	18.4	22.6	22.5	23.1	20.4	14.9	11.8	3.1		
	2%	DB	2.3	4.1	13.0	20.2	26.1	27.7	29.9	28.8	27.0	20.9	12.7	3.0		
		MCWB	0.6	1.9	8.1	10.9	15.9	19.7	22.0	21.3	19.5	14.0	9.3	1.5		
	5%	DB	0.2	2.3	9.9	17.6	22.9	26.2	27.9	27.3	23.9	17.4	9.9	2.0		
		MCWB	-0.7	0.2	6.4	9.3	14.1	18.4	20.9	20.1	18.0	12.3	6.7	0.7		
	10%	DB	-2.0	-0.2	7.2	14.8	21.1	24.0	27.2	25.2	22.3	14.1	7.3	0.1		
		MCWB	-2.9	-1.3	3.9	7.7	13.4	17.3	20.3	18.5	17.1	9.9	4.9	-0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	5.6	13.8	15.2	20.3	24.1	25.6	24.6	23.0	16.9	12.5	3.5		
		MCDB	5.5	7.7	16.8	22.3	25.5	28.7	28.8	28.5	26.6	20.2	14.6	4.6		
	2%	WB	0.8	2.2	9.0	11.9	18.0	21.8	23.8	22.7	20.7	14.8	9.5	2.0		
		MCDB	1.9	4.5	13.0	17.3	22.9	25.9	27.5	27.3	24.5	20.1	13.6	3.4		
	5%	WB	-0.8	0.1	6.0	10.2	16.1	20.1	22.5	21.3	19.3	12.8	7.1	0.6		
		MCDB	0.0	1.9	9.6	16.5	20.6	24.3	26.6	25.5	22.8	16.7	9.4	1.4		
	10%	WB	-2.6	-1.6	3.8	8.2	14.4	18.7	21.2	20.0	17.6	10.8	4.8	-0.8		
		MCDB	-2.0	-0.5	6.8	13.9	19.2	22.5	25.7	23.8	20.9	14.0	7.4	-0.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	12.7	12.3	12.4	13.5	12.8	13.3	13.5	12.6	10.3	8.7	9.2		
		MCDBR	10.0	12.4	14.2	17.8	18.4	14.9	14.7	15.3	14.7	14.8	12.6	9.2		
		MCWBR	8.6	9.6	9.4	10.3	9.7	8.2	8.1	8.6	8.7	9.0	8.5	7.7		
		MCDBR	9.6	10.7	13.0	15.6	14.1	12.1	12.7	12.7	12.2	12.9	11.5	8.7		
	5% WB	MCWBR	8.4	8.4	9.0	9.7	8.3	7.8	8.0	7.8	8.1	8.3	8.4	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.270	0.307	0.346	0.358	0.369	0.382	0.372	0.348	0.306	0.282	0.261		
	taud		2.381	2.335	2.385	2.380	2.398	2.389	2.375	2.429	2.476	2.581	2.524	2.407		
	Ebn at Noon		847	902	916	906	904	893	875	870	861	857	808	787		
	Edh at Noon		67	88	100	111	113	115	116	105	91	70	59	58		
(44) All-Sky Solar Radiation	RadAvg		1.40	2.56	3.74	4.70	5.26	5.61	6.01	5.08	3.76	2.23	1.28	0.99		
	RadStd		0.17	0.24	0.35	0.45	0.36	0.43	0.29	0.38	0.29	0.31	0.16	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page